PREG ASSIST

Project Y4 RP 25-26J-253

Project Proposal Report

B.Sc. (Hons) Degree in Information Technology Specializing in Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology

Sri Lanka

August 2025

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Declaration

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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Abstract

Maternal health emergencies remain a major cause of mortality and morbidity, especially in low-resource settings. Existing maternal education methods often lack interactivity and personalization, limiting their effectiveness. This research proposes a hybrid maternal emergency training system that integrates **Augmented Reality (AR)** and **on-device Machine Learning (ML)** to improve preparedness and response.

The system employs lightweight ML models running offline on mobile devices to analyze weekly maternal health inputs while preserving user privacy. Based on these results, it recommends **compressed AR emergency modules**, delivered from the cloud and supported with **multilingual voice guidance** in Sinhala, Tamil, and English. The approach ensures both scalability and accessibility while reducing storage requirements through model and content compression.

The development process includes requirement analysis, system design, module development, integration, and user evaluation. Expected outcomes include improved maternal knowledge retention, greater confidence in handling emergencies, and increased accessibility of training materials.

The novelty lies in combining **privacy-preserving offline ML** with **compressed cloud-based AR modules** in a maternal healthcare application, demonstrating the practical use of hybrid AR-ML systems in medical education.

Keywords: Maternal health, Augmented Reality, Machine Learning, Emergency preparedness, Privacy-preserving, Hybrid training

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1. INTRODUCTION

Maternal health remains a critical global challenge, with pregnancy-related complications representing a leading cause of death and disability among women of reproductive age. The World Health Organization (WHO) estimates approximately 295,000 maternal deaths annually, with 94% occurring in low- and middle-income countries (LMICs) [1]. The majority of these fatalities result from emergencies such as postpartum hemorrhage, hypertensive disorders, infections, and obstructed labor, where delayed recognition and insufficient emergency preparedness significantly elevate risks. In Sri Lanka and similar regional contexts, despite improvements in maternal mortality rates, gaps persist in awareness, literacy, and timely response to obstetric emergencies.

Traditional maternal education methods including pamphlets, workshops, and clinical visits often fail to provide interactive, engaging, and personalized guidance. Pregnant mothers frequently lack clear instruction, contributing to lower health literacy, increased anxiety, and weaker maternal fetal bonding. Immersive digital health interventions have demonstrated the potential to address these gaps. For instance, virtual reality (VR) interventions have been associated with reduced maternal anxiety and labor-related discomfort in approximately 89% of participants [2], while augmented reality (AR) interventions in Jakarta showed a 5-point increase in maternal knowledge and a 9.7-point improvement in healthy behavior adherence compared to conventional teaching [3].

Existing AR applications, however, demonstrate limited scope and personalization. The AR Pregnancy App focuses primarily on dietary and lifestyle guidance [4]; the Road to Birth project provides AR-based training for midwives rather than expectant mothers [5]; and widely used platforms like The Bump emphasize fetal growth tracking and general information but lack personalized emergency preparedness modules [6]. Moreover, most AR applications rely heavily on pre-downloaded content or internet connectivity, limiting accessibility and scalability for high-quality interactive experiences [9][10].

To address these gaps, this research proposes the development of a mobile-based hybrid AR and ML-powered Personalized Emergency Training Module. This system integrates:

1. AR Emergency Training Modules streamed online, comprising at least 10 3D scenarios for common or critical pregnancy emergencies (e.g., heavy bleeding, high blood pressure complications, difficult labor), embedded with multilingual voice guidance (Sinhala, Tamil, English) and interactive quizzes to reinforce

learning. The models are compressed for efficient streaming and low storage usage.

2. Offline on-device ML that analyzes weekly manually entered health data (blood pressure, pulse rate, hemoglobin levels, gestational age, symptoms, and doctor advice for the week) to classify risk levels and recommend the most relevant AR module for each user, ensuring data privacy.
3. Personalized, trimester-specific training based on individual risk profiles, allowing adaptive learning and reinforcing maternal emergency preparedness.

From an Information Technology (IT) perspective, the project demonstrates direct application of core competencies including:

1. Mobile app development optimized for performance and efficient AR streaming.
2. AR integration using Unity and ARCore/ARKit with compressed model streaming for reduced device storage usage [10][11].
3. Lightweight on-device ML deployment with TensorFlow Lite for privacy-preserving inference.
4. Data-driven personalization through rule-based and manual input processing.
5. User-centered design and accessibility engineering, ensuring cultural and linguistic adaptability in Sri Lankan contexts.

This hybrid approach combining online AR streaming with offline ML inference, interactive quizzes, and personalized scenario selection represents a technically and educationally novel solution. It advances maternal health education by providing context-aware, adaptive, and clinically relevant emergency training, filling a research and practice gap absent in existing AR pregnancy applications.

1.1 Literature Survey

This section surveys prior work on Augmented Reality (AR) in maternal health education, critically evaluates existing contributions, and identifies gaps that the proposed AR Emergency Training Module addresses. The focus is on combining immersive learning with personalized, data-driven interventions to improve maternal emergency preparedness.

1. AR-Based Pregnancy Interventions:
   Augmented Reality has increasingly been leveraged as a transformative educational tool in maternal health, providing interactive, context-aware experiences. By overlaying digital 3D models of fetal development, maternal anatomy, and simulated emergencies onto real-world environments, AR enables pregnant individuals to visualize complex physiological changes and gain practical, hands-on understanding. Critical evaluation of existing AR applications reveals several limitations. The AR Pregnancy App is primarily restricted to dietary and lifestyle guidance, offering minimal interactivity or personalization for emergency scenarios [4]. Similarly, the Road to Birth project, while providing immersive training for midwives and healthcare trainees, neglects the self-education needs of pregnant women themselves [5]. Widely adopted platforms such as The Bump focus on fetal growth tracking and general pregnancy advice but lack emergency preparedness modules and context-specific guidance [6]. Systematic reviews also indicate a scarcity of AR-focused maternal health interventions, despite evidence supporting VR's efficacy in reducing anxiety and improving knowledge retention during pregnancy [2][7]. From this analysis, it is clear that existing interventions do not synthesize personalization, adaptive risk assessment, and immersive AR in a unified framework, representing a critical gap that the proposed module aims to fill

2. Effectiveness of AR in Maternal Health Education:
   Empirical studies further underscore AR's capacity to enhance knowledge acquisition, engagement, and behavioral outcomes. In Jakarta, an AR-based maternal health education program improved participants' knowledge scores by 5.0 points and healthy lifestyle behaviors by 9.7 points relative to conventional teaching methods [3]. Other studies confirm that AR's integration with multimodal inputs such as voice guidance and interactive 3D models increases engagement and retention of complex maternal health concepts [9]. VR-based interventions have similarly been shown to mitigate labor-related anxiety and pain, suggesting that AR could replicate these outcomes with the added benefit

of interactive visualization [8]. Beyond maternal health, empirical work in other educational domains demonstrates that AR and VR can significantly enhance comprehension and engagement in complex subjects, including STEM learning [16]. Evaluation of these studies highlights a consistent strength: immersive technologies improve comprehension and engagement. However, none systematically couple risk-adaptive personalization with emergency preparedness, leaving a significant gap in the application of AR for proactive, context-sensitive maternal health training.

3. Gaps in Current AR Applications:
   Despite the proven educational benefits, current AR solutions demonstrate critical shortcomings. Most applications offer generic educational content rather than scenario-specific guidance, and adaptation based on trimester, weekly health data, or risk factors is largely absent. While some apps function offline, high-quality AR experiences often require pre-downloaded assets or constant internet access, limiting scalability and accessibility [10][11]. Furthermore, current systems rarely incorporate machine learning for context-aware scenario recommendation or adaptive training. These gaps highlight the pressing need for integrated, adaptive, and interactive AR systems that combine immersive learning with real-time, personalized guidance.

4. Justification for the Proposed AR Component:
   The proposed AR Emergency Training Module directly addresses these limitations through hybrid architecture. High-quality, compressed AR scenarios are streamed from the cloud, enabling efficient storage usage while supporting immersive 3D visualization. A lightweight TensorFlow Lite machine learning model analyzes manually entered weekly health data including blood pressure, pulse, hemoglobin levels, gestational age, symptoms, and doctor advice to classify risk and recommend the most relevant AR scenario. This on-device, offline processing ensures privacy and enables real-time, context-aware personalization. Embedded quizzes reinforce learning and guide progression within each AR scenario. Multilingual guidance in Sinhala, Tamil, and English enhances accessibility for diverse populations. By combining these elements, the system provides personalized, trimester-specific emergency preparedness, overcoming the limitations of existing AR interventions. This approach synthesizes immersive visualization, adaptive machine learning, and educational scaffolding in a coherent, learner-centered framework.

5. Integration of Machine Learning in AR Pregnancy Education:

While traditional AR pregnancy apps primarily focus on visualization or static information delivery, the proposed module incorporates intelligent, offline personalization. Using manually entered weekly health inputs, the ML component dynamically selects relevant emergency scenarios, ensuring adaptive and clinically relevant training. This integration enables personalized emergency preparedness tailored to individual risk profiles, privacy-preserving on-device operation, and adaptive learning through quizzes and dynamic scenario selection. Evaluative synthesis demonstrates that the proposed approach fills a critical gap in the literature, as no existing AR application combines cloud-based AR streaming, offline ML personalization, multilingual guidance, quizzes, and adaptive emergency preparedness in a single system [3-11].

6. AR Asset Optimization and ML-Driven Personalization in Maternal Health:
   Augmented Reality applications in maternal health are increasingly recognized for enhancing visualization and engagement; however, empirical studies on asset streaming, compression, and optimization remain scarce. Compression evaluations in AR domains, including light field [12] and digital twin assets [13], demonstrate that lossless and lightweight compression can significantly reduce loading latency and memory usage while maintaining perceptual fidelity on mobile devices. These findings suggest that AR pregnancy systems should move beyond pre-bundled assets toward streamed, optimized models. Parallelly, scoping reviews of machine learning for maternal risk prediction reveal that while many models address hypertensive disorders, gestational diabetes, and hemorrhage, few integrate these predictions with adaptive, scenario-specific AR training [14]. Further formative studies show that pregnant users value personalized guidance, interactive learning, and content that adapts to health status and risk factors [15]. Taken together, these insights indicate a critical opportunity to synthesize AR streaming, model compression, and ML-driven personalization in a unified, mobile-based maternal emergency training module.

1.2 Research Gap

Despite the increasing use of augmented reality in maternal health, current applications remain limited in delivering adaptive, personalized emergency preparedness for pregnant women. Existing platforms predominantly provide static content, such as dietary guidance, fetal growth visualization, or professional training for midwives, and lack context-aware, risk-stratified, or individualized interventions. These applications demonstrate minimal intelligent adaptability, depend heavily on pre-downloaded or fixed AR content, and rarely incorporate predictive analytics or machine learning to guide scenario selection. Additionally, immersive AR systems often require substantial device storage or continuous internet connectivity and seldom utilize model compression or streaming techniques for optimized mobile deployment.

Current literature also highlights gaps in the integration of on-device intelligence for personalized health assessment. Risk stratification based on weekly pregnancy data including vital signs, gestational age, symptoms, and medical advice remains largely unexplored. Moreover, there is limited research on leveraging cross-domain transfer learning to adapt models trained on general health datasets to the maternal health domain, which could enhance predictive accuracy and reduce dependency on extensive labeled pregnancy data.

Accessibility and multimodal data acquisition are also underdeveloped. Few systems accommodate varying literacy levels or provide multilingual support, and there is little evidence of mechanisms to assess user understanding or adapt training content dynamically. Overall, existing AR interventions fail to combine scalable visualization, privacy-preserving computation, intelligent personalization, and dynamic adaptation in a single framework, leaving a significant technical and educational void in maternal emergency preparedness technology.

1.3 Research Problem

Current augmented reality applications in maternal health predominantly offer generalized guidance, lacking personalization and dynamic adaptability. Pregnant women often receive static or pre-determined AR content that does not adjust to their evolving weekly health data, symptoms, or risk factors. This limitation prevents the delivery of interactive, context-aware support during pregnancy emergencies, reducing engagement and the practical effectiveness of AR training.

Existing solutions also rarely integrate lightweight on-device machine learning or employ cross-domain transfer learning techniques to enhance predictive accuracy and improve personalized scenario selection. Consequently, AR applications cannot reliably prioritize emergency scenarios relevant to an individual's health profile, gestational stage, or previous exposure to training modules. Furthermore, the absence of predictive intelligence restricts real-time adaptation and undermines the potential for AR to deliver targeted, clinically meaningful emergency preparedness.

Another critical challenge lies in the insufficient alignment of educational theory with immersive technology. Many AR interventions emphasize visualization and simulation without embedding formative assessment, adaptive learning strategies, or feedback mechanisms to reinforce knowledge retention and emergency response skills. This gap limits the ability of expectant mothers to translate AR training into confident, informed decision-making during real-world emergencies.

Finally, current platforms lack continuous context-aware evaluation of user comprehension or progress. Without integrated feedback, assessment, and dynamic content adaptation, AR-based maternal health education remains underutilized and fails to achieve its full potential as a personalized, evidence-based training tool. These deficiencies highlight the pressing need for a system that unites adaptive AR visualization, cross-domain machine learning personalization, and pedagogically grounded design to support actionable emergency preparedness for pregnant women.

2. OBJECTIVES

2.1 Main Objective

The main objective of this research is to design and implement a Personalized AR Emergency Training Module for maternal health that delivers adaptive, context-aware, and interactive training for pregnant women. The system leverages lightweight, on-device machine learning models to analyze weekly health inputs and employs cross-domain transfer learning to enhance model accuracy and adaptability. Using compressed, cloud-streamed AR modules with multilingual guidance (Sinhala, Tamil, and English), the solution provides trimester-specific, personalized emergency training while preserving user privacy. The module is designed to support formative assessment through embedded quizzes and dynamic scenario adaptation, thereby promoting actionable learning, engagement, and emergency preparedness.

2.2 Specific Objectives

1. Develop a cloud-streamed AR-based training environment simulating critical maternal health emergencies, with step-by-step guidance and embedded formative assessments.

2. Integrate lightweight, on-device machine learning models to analyze weekly pregnancy data and recommend personalized AR training scenarios.

3. Employ cross-domain transfer learning to leverage pre-trained general health models for pregnancy-specific emergency risk classification, improving training efficiency and model adaptability.

4. Incorporate pedagogically informed features, such as embedded quizzes and scenario progression based on user comprehension, to enhance retention, skill acquisition, and confidence in emergency response.

5. Ensure multilingual accessibility with voice narration in Sinhala, Tamil, and English to accommodate diverse literacy levels and cultural contexts.

6. Evaluate the system's effectiveness in improving user engagement, learning outcomes, and emergency preparedness, comparing performance against existing AR-based maternal health applications.

3. METHODOLOGY

3.1 Overall System Diagram

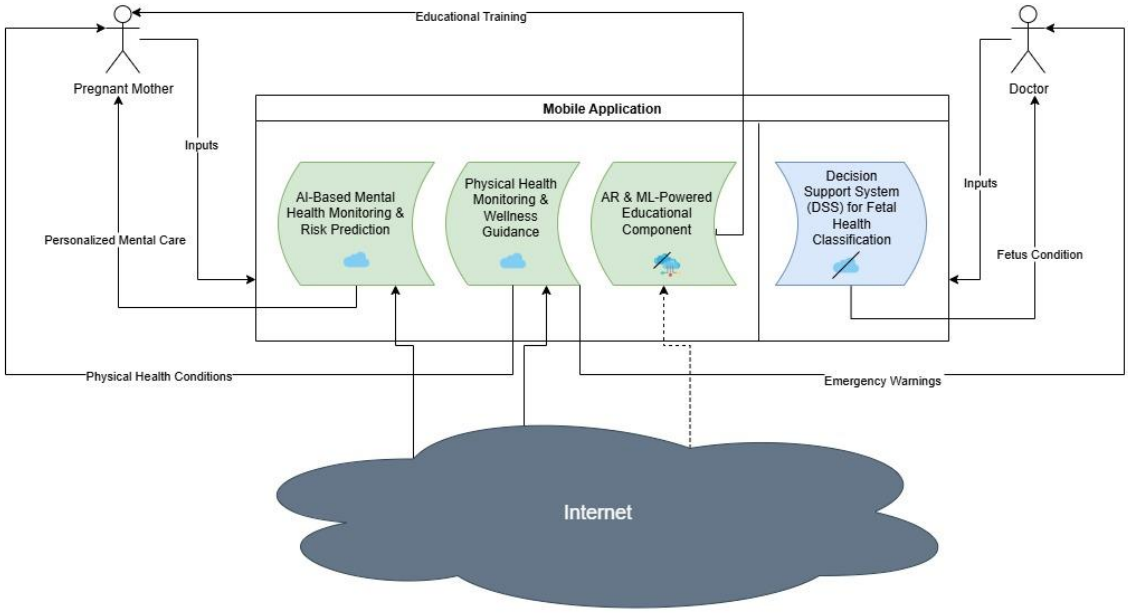

Figure 1: Overall System Diagram

The system depicted in the diagram is Preg Assist, a mobile application designed to support pregnant women throughout their pregnancy journey.

For Pregnant Mothers:

1. AI-Based Mental Health Monitoring & Risk Prediction: A fully online module that uses cloud-hosted machine learning models to monitor mental health and predict potential risks.
2. Physical Health Monitoring & Wellness Guidance: Also fully online, this module continuously tracks physical health and provides personalized wellness guidance through cloud connectivity.
3. AR & ML-Powered Educational Component: A hybrid module combining offline, locally run machine learning for personalized risk assessment with cloud-based augmented reality features to deliver interactive educational training.

For Doctors:

1. Decision Support System (DSS) for Fetal Health Classification: A fully offline module that runs locally on mobile devices, enabling doctors to classify fetal health and receive emergency alerts without relying on internet access.

By leveraging both mobile and cloud technologies, Preg Assist ensures personalized, continuous care and education while remaining accessible and reliable in varying connectivity conditions. It also facilitates secure interaction between pregnant mothers and doctors through data sharing and alerts, promoting healthier pregnancy outcomes.

3.2 Individual System Diagram

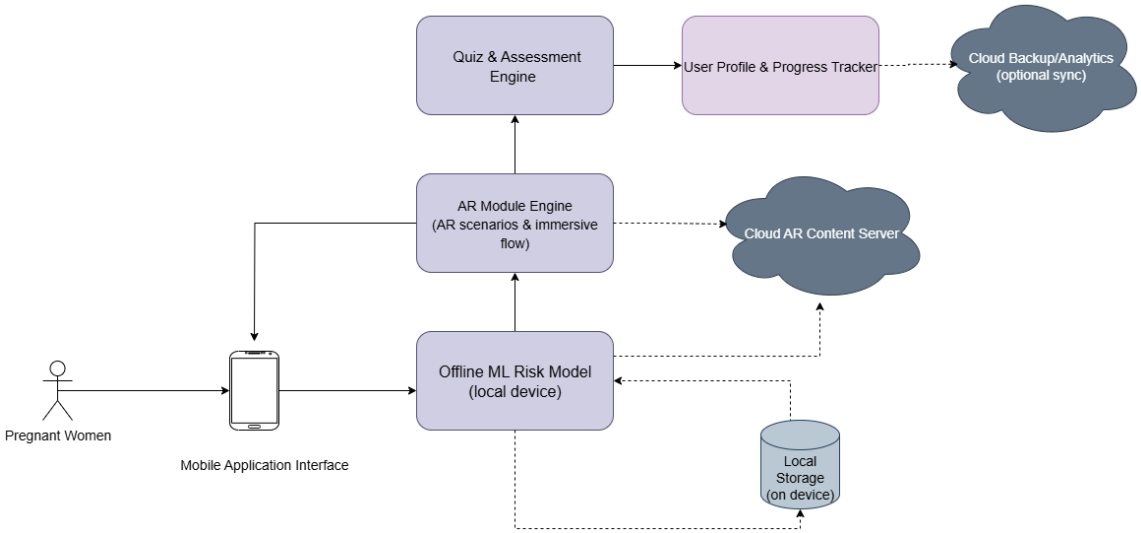

Figure 2: Individual System Diagram

The diagram illustrates the core components of the AR-based maternal emergency training module and their interactions. Pregnant women and healthcare professionals are the primary actors, interacting with the Mobile Application Interface, which serves as the main entry point to the system.
Inside the module:

1. The AR Module Engine delivers immersive training scenarios and now communicates with the Quiz & Assessment Engine to provide embedded decision-based quizzes.
2. The Offline ML Risk Assessment personalizes the AR scenarios based on user-specific data, enhancing the learning experience.

3. The User Profile & Progress Tracker records quiz results, tracks performance, and maintains a personalized learning path.

External clouds support the system:

1. The Cloud AR Content Server provides AR assets to the AR Module Engine for high-fidelity experiences.
2. Cloud Backup/Analytics optionally stores user progress for remote monitoring or backup.

This architecture ensures that training is personalized, interactive, and scalable, while allowing offline functionality and seamless content updates when online.

3.3 Project Approach

The project follows a design and development approach that combines Augmented Reality (AR), Machine Learning (ML), and pedagogical strategies to create an interactive, personalized, and context-aware maternal emergency training system. The development process is iterative, ensuring continuous refinement based on testing and user feedback. The approach emphasizes both technical performance and educational effectiveness, aligned with Bloom's taxonomy focusing on Create and Evaluate.

The development process consists of five stages:

3.3.1 Tasks and Sub-Tasks

Task 1: Requirement Gathering

1. Identify user needs for pregnant women and healthcare professionals.
2. Define functional and non-functional requirements, including AR streaming, ML personalization, quizzes, and multilingual accessibility.

Task 2: System Design

1. Design AR simulations and training scenarios representing maternal health emergencies.
2. Plan system architecture, UI/UX, and integration of offline ML models with cloud-streamed AR modules.

Task 3: Module Development

1. ML Model Development: Train lightweight, privacy-preserving TensorFlow Lite models for personalized risk assessment and scenario selection, leveraging cross-domain transfer learning.

2. 3D Models Development: Create high-quality, compressed AR-ready 3D models and animations for maternal emergencies, optimized for streaming and mobile deployment.
3. Mobile App Development: Build the mobile application using Flutter and Dart, integrating AR streaming, offline ML, quizzes, and progress tracking.
4. Additional Features: Implement interactive quizzes embedded in AR scenarios, multilingual guidance (Sinhala, Tamil, English), and dynamic content adaptation based on weekly health data.

Task 4: Integration and Testing

1. Integrate AR simulations, ML models, quizzes, and mobile app components.
2. Conduct functional, usability, performance, and accessibility testing, ensuring smooth AR streaming and accurate offline ML inference.

Task 5: User Evaluation

1. Collect feedback from pregnant women and healthcare professionals.
2. Evaluate system effectiveness in improving knowledge retention, emergency preparedness, and user engagement.
3. Analyze AR module interactions, quiz performance, and personalized scenario selection to refine the system.

3.3.2 Materials and Tools

Equipment

1. Mobile devices (Android/iOS)
2. AR-enabled hardware (smartphones/tablets)

Software and Frameworks

1. AR Development: Unity 3D, ARCore/ARKit
2. Mobile App Development: Android Studio, VS Code, Flutter, Dart
3. Machine Learning: TensorFlow Lite, Python
1. Design Tools: Figma or Adobe XD for UI/UX

Other Resources

2. Datasets from Kaggle and Hugging Face for general health and pregnancy-related ML training
3. Research articles, maternal health guidelines, and emergency protocols from IEEE, Springer, and PubMed

3.3.3 Data Collection

Primary Data

1. User performance metrics, quiz results, and interaction logs during AR training sessions
2. Feedback on usability, clarity, and confidence-building

Secondary Data

1. Existing maternal health protocols and guidelines
2. Published research on AR interventions, maternal health, and ML-based personalization

Anticipated Conclusion

The project aims to show that a personalized, AR-based training system can improve knowledge retention and decision-making in maternal emergencies. By combining offline machine learning for risk assessment, cloud-streamed AR modules for immersive learning, and embedded quizzes for continuous assessment, it enhances preparedness and confidence among pregnant women and healthcare professionals. The mobile-based, scalable system is adaptable to diverse users and evolving training scenarios.

4. PROJECT REQUIREMENTS

4.1 Functional Requirements

These requirements define the essential functional and technical features of the AR Emergency Training Module, emphasizing personalized AR training, machine learning-driven scenario selection, and immersive, interactive maternal health education.

1. **AR-Based Simulation**: The system shall provide immersive, cloud-streamed AR simulations of critical maternal health emergencies. Each module should be compressed for efficient mobile deployment and include step-by-step guidance through the scenario.

2. **Personalized Scenario Selection**: The system shall adapt training content based on user-reported weekly health data, including parameters such as blood pressure, pulse, hemoglobin levels, gestational age, symptoms, and doctor advice. Scenario recommendations shall reflect trimester-specific and context-aware emergency training needs.

3. **Machine Learning Integration**: The system shall incorporate lightweight, on-device ML models to analyze weekly pregnancy inputs, classify risk levels, and recommend the most relevant AR scenarios. All processing shall occur locally to preserve privacy and ensure offline inference capability.

4. **Cross-Domain Transfer Learning**: The system shall leverage pre-trained general health models and adapt them for pregnancy-specific emergency risk classification, improving predictive accuracy and reducing reliance on extensive labeled pregnancy datasets.

5. **Interactive Feedback and Formative Assessment**: The system shall embed quizzes within AR scenarios to assess comprehension, provide immediate context-aware feedback, and guide progression through the training modules.

6. **Multilingual Support**: The system shall offer voice narration and guidance in Sinhala, Tamil, and English to accommodate varying literacy levels and cultural contexts.

7. **Progress Tracking**: The system shall maintain records of completed AR scenarios, quiz performance, and personalized scenario recommendations to monitor learning outcomes and training engagement over time.

8. Hybrid Architecture: The system shall operate with ML processing offline for privacy while streaming compressed AR modules online for high-quality visualization without excessive storage requirements.

4.2 Non-Functional Requirements

1. **Usability**: The system shall provide an intuitive, user-friendly interface suitable for pregnant women of varying literacy levels as well as healthcare professionals, with clear navigation, accessible controls, and multilingual support (Sinhala, Tamil, English).

2. **Performance**: AR simulations shall run smoothly with minimal latency on standard mobile devices. Compressed AR models and optimized streaming shall ensure high-quality visualization without excessive load times or performance degradation.

3. **Reliability**: The system shall deliver consistent, accurate, and context-aware guidance during all training scenarios, including personalized scenario selection via the on-device ML component.

4. **Scalability**: The system shall support multiple concurrent users, sessions, and cloud-streamed AR modules without impacting performance or accessibility.

5. **Security & Privacy**: The system shall protect all user data. On-device ML ensures privacy by processing sensitive health inputs locally, while secure cloud storage and streaming protocols prevent unauthorized access to AR content.

6. **Portability**: The system shall be deployable on widely available mobile devices with AR capabilities, supporting Android and iOS platforms and standard AR hardware.

7. **Maintainability**: The system shall allow for efficient updates, bug fixes, AR module additions, and ML model retraining, including adaptation of pre-trained models via cross-domain transfer learning.

8. **Accessibility & Inclusivity**: The system shall accommodate users with varying literacy, language, and cultural backgrounds through multilingual guidance, clear visual cues, and simplified manual data entry forms.

9. **Offline Functionality**: While AR modules are cloud-streamed, all ML computations and user health data processing shall occur offline to ensure privacy and continued functionality regardless of connectivity.

5. COMMERCIALIZATION

Table 1: Budget Expenditure

Description	Occurrence	Cost
AR Model Assets	One-Time	LKR 2,000 and up (depends on asset complexity)
Model Training	Pay-As-You-Go	LKR 3,000 per GPU unit
3D Model Creation & Animation	One-Time	LKR 5,000–15,000 per model
Software Licenses (Unity, AR SDK)	Annual/One-Time	LKR 10,000–30,000
Cloud Databse	Pay-As-You-Go	0.026 USD per GB

5.1 Commercialization Methods

Currently, there is no dedicated AR-based personalized training solution for maternal emergency preparedness. Existing tools are limited to general education or static simulations, creating an opportunity to introduce a more interactive, adaptive, and scalable system. The cloud-streamed AR modules and offline ML personalization ensure that the system can be easily updated and scaled for multiple users. To ensure both accessibility and sustainability, two key commercialization methods are proposed:

1. Freemium Model – Provide a basic free version for individuals (pregnant women, students) with optional paid upgrades for advanced features, premium AR scenarios, and personalized learning modules.

2. Partnerships with Health Organizations – Collaborate with NGOs, government health agencies, and hospitals to integrate the system into maternal health programs through bulk licensing or sponsored deployments.

6. FAILURE MODE ANALYSIS

Table 2: Failure Mode Analysis

Failure Mode	Impact	Severity/ Priority	Mitigation Strategy
ML Personalization Fails: ML models provide inaccurate guidance (e.g., misclassifying user level).	Irrelevant or ineffective training; reduced learning effectiveness.	**High**	Provide default standardized modules; allow manual user input for knowledge level; retrain models continuously.
AR Simulation Performance Issues: AR modules lag, crash, or fail on low-end devices.	Frustration, reduced engagement, or abandonment of training.	**High**	Optimize and compress AR assets; provide "Lite Mode" with simplified 3D assets; enable cloud-streamed AR modules.
Cross-Domain Transfer Learning Fails: Transfer learning does not adapt well to maternal health domain.	Inaccurate predictions and weak personalization; reduced credibility.	**Medium**	Use maternal-health-specific fallback datasets; apply hybrid rule-based + ML approach; fine-tune with feedback.
Real-Time Feedback Malfunctions: Feedback is delayed, missing, or incorrect.	Users may receive misleading instructions; loss of trust.	**High**	Provide basic instructional prompts as backup; pre-cache essential guidance; log errors and refine models.
Data Privacy and Security Breach: Weak security exposes user data.	Loss of trust, ethical concerns, legal issues.	**High (Critical)**	Apply end-to-end encryption; prioritize local-only storage; comply with GDPR/health data standards.
System Integration Failure: AR, ML, and app modules do not integrate smoothly.	Inconsistent performance; incomplete user experience.	**High**	Use modular architecture; conduct early integration testing; maintain backup standalone modules for partial use.

7. GANTT CHART

Table 3: Gantt Chart

Tasks	Sep	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May
Collect and preprocess maternal health datasets									
Train baseline ML models for personalization/prediction.									
Compress ML models									
Test performance on mobile devices.									
Define maternal emergency simulations									
Storyboard scenarios with multilingual narration.									
Develop 3D assets and animations.									
Integrate into Unity + ARCore/ARKit.									
Optimize for mobile performance.									
Flutter/Dart app development									
Integrate ML personalization + AR scenarios.									
Add features: interactive feedback, quizzes, progress tracking.									
Functional testing against requirements.									
Performance testing for smooth AR + ML inference									
Conduct pilot study with pregnant women and healthcare professionals.									
Collect feedback									
Refine AR modules, ML personalization, and UI/UX.									
Final documentation + commercialization strategy									

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